PRESENTATION

The Future of AI + Blockchain

Historical foundations, current state, protocol stack, and the path to an agentic financial system

01 Past

- Digital money and internet payments
- Bitcoin, Ethereum, DeFi, stablecoins
- AI from automation to agents

02 Present

- Convergence of AI and blockchain
- Agent commerce and programmable payments
- Protocols: x402, MPP, AP2, A2A, MCP

03 Future

- Machine-to-machine economy
- Autonomous financial coordination
- Agentic Payments Bot as a practical blueprint

AI + BLOCKCHAIN • AGENTIC COMMERCE • MACHINE ECONOMY

HISTORICAL RETROSPECTIVE

From Digital Money to Programmable Value

01
Pre-internet finance
Cash, ledgers, correspondent banking

02
1990s–2000s
Card networks, PayPal, e-commerce payments

03
2009
Bitcoin: digital scarcity and peer-to-peer value

04
2015
Ethereum: smart contracts and programmable settlement

05
2020–2024
DeFi, tokenization, stablecoins, onchain liquidity

06
2025+
Agentic commerce begins to emerge

Why blockchain changed finance

- Native digital assets
- Programmable trust and settlement
- Composability across applications
- Global, 24/7, API-like money rails

AI's Path: From Automation to Autonomous Agents

01
Expert systems
Rule-based automation

02
Machine learning
Prediction from data

03
Deep learning
Perception and pattern recognition

04
Transformers
Language, reasoning, code

05
LLM agents
Tool use, planning, workflows

06
Multi-agent systems
Coordination, negotiation, execution

What AI adds to finance

- Intent understanding
- Dynamic planning
- Decision support at scale
- Autonomous execution through tools
- Continuous optimization and negotiation

The convergence thesis

Blockchain supplies programmable value; AI supplies programmable decision-making.

Where AI and Blockchain Meet Today

AI is becoming agentic

- LLMs now use tools, APIs, memory, and workflows
- Agents can search, compare, negotiate, and execute
- Multi-agent systems are moving from demos to products
- Need: trusted actions, budgets, permissions, observability

Blockchain is becoming usable infrastructure

- Stablecoins make internet-native settlement practical
- Wallets become machine accounts and programmable treasuries
- DeFi and tokenization expand financial functionality
- Need: compliance, identity, UX, and interoperability

Convergence in practice

- Agentic payments
- API monetization
- Onchain coordination
- Machine-to-machine commerce

The Emerging Stack for Agentic Commerce

Applications and Skills	Agents, bots, assistants, marketplaces
Coordination	A2A: agent-to-agent messaging, delegation, negotiation
Payment Protocols	x402, MPP, AP2
Context and Tooling	MCP: model context protocol for tools, data, and external systems
Settlement Rails	Stablecoins, smart contracts, bank rails, cards, custodial and non-custodial wallets

How to read the stack

- MCP is not a payment rail
- A2A handles inter-agent coordination
- x402, MPP, and AP2 handle payment interaction patterns
- Rails settle value in crypto or traditional finance

What Each Protocol Does

	DEFINITION	BEST USE CASE
x402	HTTP-native payment handshake for paid resources and API access	Micropayments, pay-per-call data and compute
MPP	Machine Payments Protocol for machine-to-machine payment orchestration across payment methods	Streaming or recurring machine payments, broader payment abstraction
AP2	Mandate-based agentic payment flow with delegated authorization	Agent purchases with policies, credentials, and approval controls
A2A	Agent-to-agent communication and task delegation layer	Negotiation, service discovery, multi-agent workflows
MCP	Universal context and tool interface for models and agents	Connecting external data sources, tools, apps, and other agents

Together, these protocols turn agents from chat interfaces into transacting economic actors.

PROJECT SPOTLIGHT

Agentic Payments Bot / Skill / Assistant

A dual-protocol agentic payment service for the Open Agent Skills Ecosystem.
It supports x402 and AP2, bridges web3 and web2 payment rails, and adds policy, security, and audit controls for autonomous payments.

01 What it is

- TypeScript payment service, gateway, bot, and skill
- Implements x402 client/server and AP2 client/server
- Designed for AI agents, assistants, and open agent skills

02 What it supports

- Web3 via Ethereum tooling and stablecoin flows
- Web2 gateways: Stripe, PayPal, Visa Direct, Mastercard Send
- Also supports Google Pay and Apple Pay backends

03 Why it matters

- Bridges crypto-native and traditional payment worlds
- Adds governance, policy, and human confirmation
- Offers a concrete blueprint for agentic finance

The repo focuses especially on x402 and AP2, while fitting into the larger ecosystem with A2A and MCP.

Inside the Agentic Payments Bot Architecture

INPUT LAYER

- Agent or user
- CLI
- Web API
- Open Agent Skills integration

ORCHESTRATION LAYER

- Payment Intent parser
- Orchestrator
- Protocol router
- Policy engine
- Human confirmation loop

PROTOCOL & EXECUTION LAYER

- x402 client/server
- AP2 client/server
- Payment backends

Ethereum / stablecoins | Stripe | PayPal | Visa Direct | Mastecrard Send | Google Pay | Apple Pay

SECURITY, OBSERVABILITY & RELIABILITY

- KMS providers: AWS KMS, OS keyring, D-Bus, GPG, Local AES
- SQLite audit log
- Winston logging
- Dry-run mode

Core idea: parse payment intent, enforce policy, route by protocol, execute on the best rail, log everything.

Operational Principles and Payment Flows

01 Intent
Agent or user expresses a payment goal

02 Parse
Convert output into a structured PaymentIntent

03 Govern
Check rules, budgets, recipients, and thresholds

04 Route
Choose x402, AP2, or a direct gateway/backend

05 Execute
Sign, credential, pay, confirm, and receive result

06 Record
Write audit trail and return transaction state

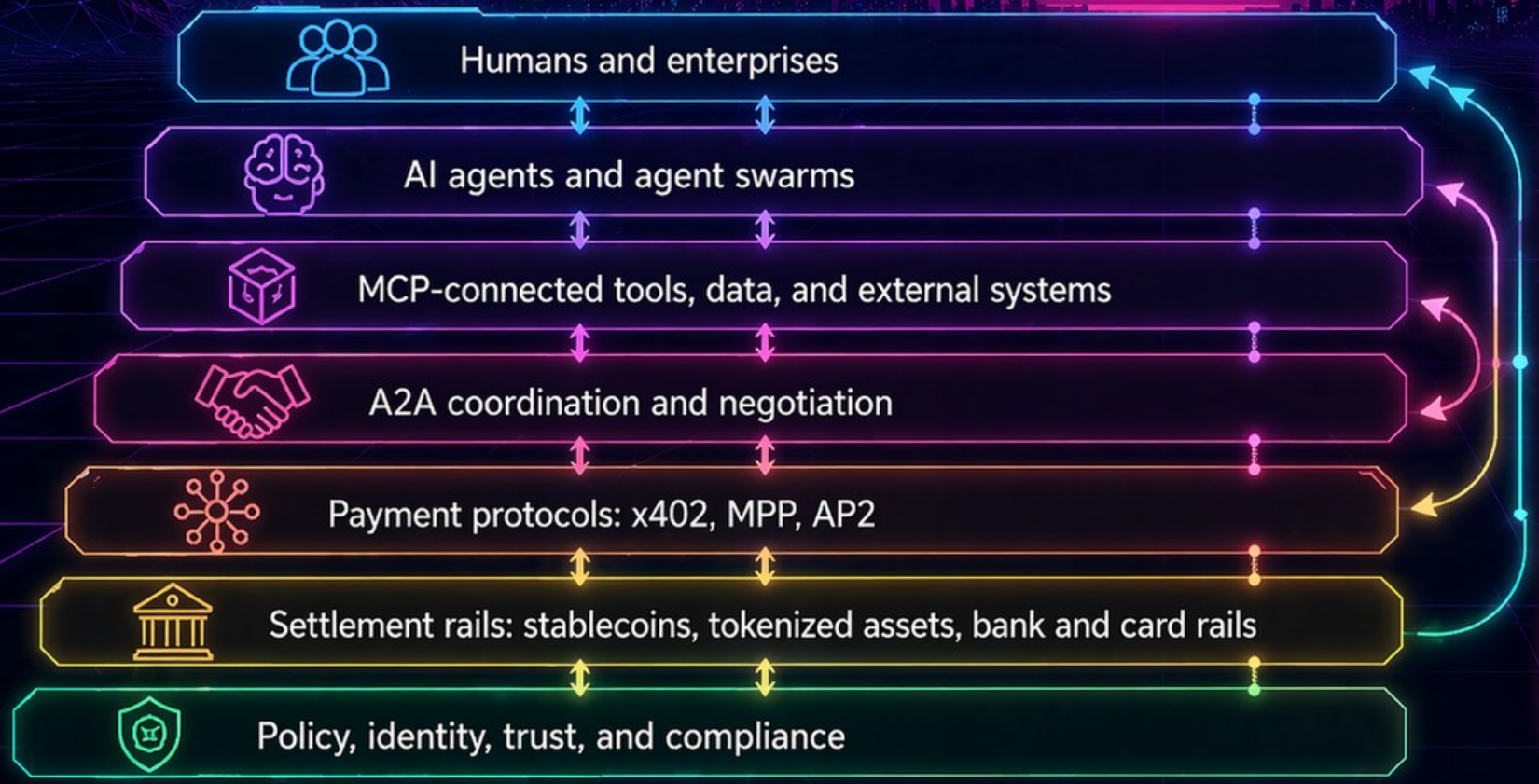
Design principles

- Protocol modularity
- Hybrid web2 + web3 support
- Secure key handling
- Human-in-the-loop when needed
- Compliance and observability by default

x402 flow: discover → sign → pay → receive

AP2 flow: create mandate → sign → get credentials → submit

Toward an Agentic Financial and Economic System



WHAT THIS FUTURE LOOKS LIKE

- Machine wallets and programmable treasuries
- Autonomous purchasing and selling
- Real-time global settlement
- Tokenized services and digital labor markets
- Finance that becomes API-native and agent-native

The long-term vision: AI becomes the decision layer; blockchain becomes the value layer.